



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2014
YEAR 11
Yearly Examination

Mathematics

General Instruction

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen.
Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Each Section is to be returned in a separate bundle.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question.
- Answer in simplest exact form unless otherwise instructed

Total Marks – 75

- Attempt questions 1-6
- Hand up in 2 sections clearly marked A & B

Examiner: *T.Evans*

Question 1**12 Marks**

- a) Determine whether the point $(-1, -8)$ lies on the line $2x - 3y - 23 = 0$ 2
- b) Solve $2^x = 3$ 2
- c) Find the value of $\left(\frac{-27}{8}\right)^{\frac{-4}{3}}$ 1
- d) Given that $\log_a b = 2.75$ and $\log_a c = 0.25$, find the value of: 2
- i) $\log_a \left(\frac{b}{c}\right)$ 2
- ii) $\log_a (bc)^2$
- e) State whether the following are ODD, EVEN or neither: 1
- i) $f(x) = 3x + 1$ 1
- ii) $f(x) = x^4 - x^2$
- f) The first term of geometric sequence is 2 and the fourth term is 128. Find the common ratio. 1

Question 2**13 Marks**

- a) For the following sequence: 36, 24, 16,..... 2
- Determine whether it is an Arithmetic or Geometric Progression, showing reasons.
- b) Find $\lim_{x \rightarrow 4} \frac{x^2 - x - 12}{x - 4}$ 2
- c) Differentiate
- i) $4x^2 - x + 1$ 1
- ii) $\frac{3}{\sqrt[3]{x}}$ 2
- iii) $\frac{3x^5 - 2x^3 + 5}{x^2}$ 2
- iv) $\sqrt{3x^2 - 3}$ 2
- v) $\frac{3x^2}{5 - 3x}$ 2

Question 3**13 Marks**

- a)** i) Find all the x & y -intercepts for the curve with the equation $y = x^3 - 1$. 2
- ii) Neatly sketch the curve $y = x^3 - 1$, showing the information from part i). 2
- iii) Find the equation of the tangent to the curve $y = x^3 - 1$ at the point where $x = 1$. 2
- b)** If $f(x) = 15x^{-2} - 9x^3$, find the value of $f'(-1)$. 2
- c)** Differentiate $f(x) = x^2 - 3x$ by first principles using the definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ 2
- d)** Simplify $\frac{\left(-2x^{\frac{1}{2}}y^{\frac{1}{3}}\right)^2}{\left(3x^{-1}y\right)^{\frac{1}{3}}} \times \frac{4x^{\frac{1}{3}}y^{-\frac{1}{3}}}{\left(x^{-1}y\right)^{\frac{1}{3}}}$ 3

Question 4**12 Marks**

a) The equation $2x^2 + 5x - 1 = 0$ has roots α and β . Find the values of:

- | | |
|---|---|
| i) $\alpha + \beta$ | 1 |
| ii) $\alpha\beta$ | 1 |
| iii) $\frac{1}{\alpha} + \frac{1}{\beta}$ | 2 |
| iv) $\alpha^2 + \beta^2$ | 2 |

b) Consider the parabola $4y = x^2 - 4x$

- | | |
|--|---|
| i) Show, algebraically, how the parabola can be expressed in the form $(x - 2)^2 = 4(y + 1)$ | 2 |
| ii) Write down the coordinates of the vertex and the focus. | 2 |
| iii) Find the equation of the directrix. | 2 |

Question Five**13 Marks**

- a)** i) Sketch $y = \log_2(x + 2)$, showing the x & y -intersections and the asymptote. 2
- ii) Hence, state the domain and range of $y = \log_2(x + 2)$. 2
- b)** Find A , B & C such that $A(x - 1)^2 + Bx + c = x^2$ 3
- c)** In a certain arithmetic series the 9th term is 78, and the 17th term is 50. Find:
- i) The first term and the common difference 2
- ii) The sum of the first forty terms 2
- iii) The value of n for which the sum of the series is first negative. 2

Question 6**12 Marks**

a) If $f(x) = x^2 - x$ find the simplest expression for $f(x + h)$ 2

b) Express $0.42 + 0.0042 + 0.000042 + \dots$ as a simplified fraction 2

c) A function $y = f(x)$ is defined as follows:

$$f(x) = \begin{cases} -2, & x \leq -1 \\ x + 1, & -1 < x < 2 \\ x^2 + 1, & x \geq 2 \end{cases}$$

i) Evaluate $f(-1) + f(2)$ 2

ii) Write an expression for $f(a^2 + 2)$ 2

d) Solve for x

i) $4^x = 12(2)^x - 32$ 2

ii) $2\log_5 3 = \log_5 x - \log_5 6$ 2

End of Examination

2014 Year 11 Mathematics Yearly - Solutions

Question 1

(a) $2x - 3y - 23 = 0$ $(-1, -8)$
 $2(-1) - 3(-8) - 23 = -1 \quad (\neq 0)$
 $(-1, -8)$ not on line

(b) $2^x = 3$
 $\log 2^x = \log 3$
 $x \log 2 = \log 3$
 $x = \frac{\log 3}{\log 2} \approx 1.585 \quad (3 \text{ d.p.})$

(c) $\left(-\frac{27}{8}\right)^{-\frac{4}{3}} = \left(-\frac{3}{2}\right)^{-4} = \frac{16}{81}$

(d) (i) $\log_a\left(\frac{b}{c}\right) = \log_a b - \log_a c = 2.5$
(ii) $\log_a (bc)^2 = 2(\log_a b + \log_a c)$
 $= 6$

(e) (i) $f(x) = (3x+1)$ $f(-x) = -3x+1$
neither
(ii) $f(x) = x^4 - x^2$, $f(-x) = x^4 - x^2$
even.

(f) $a = 2$, $T_4 = 128$
 $ar^3 = 128$
 $r^3 = 64$
 $r = 4$

Question 2

$$(a) \quad \frac{24}{36} = \frac{16}{24} = \frac{2}{3}$$

$\therefore$ geometric

$$(b) \quad \lim_{x \rightarrow 4} \frac{x^2 - x - 12}{x - 4} = \lim_{x \rightarrow 4} \frac{(x-4)(x+3)}{x-4}$$
$$= \lim_{x \rightarrow 4} x + 3 = 7.$$

$$(c)(i) \quad f(x) = 4x^2 - x + 1 \quad f'(x) = 8x - 1$$

$$(ii) \quad f(x) = 3x^{-\frac{1}{3}}, \quad f'(x) = -x^{-\frac{4}{3}}$$
$$= -\frac{1}{\sqrt[3]{2^4}}$$

$$(iii) \quad f(x) = \frac{3x^5 - 2x^3 + 5}{x^2} = 3x^3 - 2x + 5x^{-2}$$

$$f'(x) = 9x^2 - 2 - \frac{10}{x^3}$$

$$(iv) \quad f(x) = (3x^2 - 3)^{\frac{1}{2}} \quad f'(x) = \frac{1}{2}(3x^2 - 3)^{-\frac{1}{2}} \times 6x$$
$$= \frac{3x}{\sqrt{3x^2 - 3}}$$

$$(v) \quad f(x) = \frac{3x^2}{5-3x} = \frac{u}{v} \quad f'(x) = \frac{v u' - u v'}{v^2}$$

$$f'(x) = \frac{(5-3x) \times 6x - 3x^2(-3)}{(5-3x)^2}$$

$$= \frac{30x - 18x^2 + 9x^2}{(5-3x)^2}$$

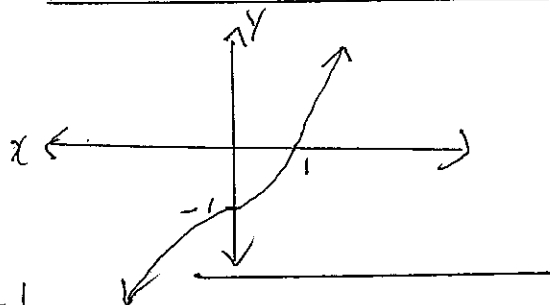
$$= \frac{30x - 9x^2}{(5-3x)^2}$$

Question 3

(a)(i) $y = x^3 - 1$

$y = 0 \quad x = 1, (1, 0), \quad x = 0 \quad y = -1 \quad (0, -1)$

(a)(ii)



(iii) $y = x^3 - 1$

$y' = 3x^2, \quad x = 1 \quad y' = 3$

$y - y_1 = m(x - x_1) \quad (1, 0)$

$y - 0 = 3(x - 1) \quad , \quad y = 3x - 3$

(b) $f(x) = 15x^{-2} - 9x^3, \quad f'(x) = -30x^{-3} - 27x^2$

$f'(-1) = -30(-1) - 27(1) = 3$

(c) $\lim_{h \rightarrow 0} \frac{(x+h)^2 - (x^2 - 3x)}{h} = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 3(x+h) - (x^2 - 3x)}{h}$

$= \lim_{h \rightarrow 0} \frac{2xh + h^2 - 3h}{h}$

$= \lim_{h \rightarrow 0} 2x + h - 3 = 2x - 3$

(d) $\frac{(-2x^{\frac{1}{2}}y^{\frac{1}{3}})^2}{(3x^{-1}y)^{\frac{1}{3}}} \times \frac{4x^{\frac{1}{3}}y^{\frac{1}{3}}}{(x^{-1}y)^{\frac{1}{3}}}$

$= \frac{16x^{\frac{1}{3}}y}{3^{\frac{1}{3}}x^{-\frac{2}{3}}y^{\frac{2}{3}}}$

$= \frac{16}{\sqrt[3]{3}} x^2 y^{\frac{1}{3}}$

$$4) a) i) \alpha + \beta = -\frac{b}{a}$$

$$= -\frac{5}{2}$$

$$ii) \alpha\beta = \frac{c}{a}$$

$$= -\frac{1}{2}$$

$$iii) \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$$

$$= \frac{\left(-\frac{5}{2}\right)}{\left(-\frac{1}{2}\right)}$$

$$= 5$$

$$iv) \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= \left(-\frac{5}{2}\right)^2 - 2\left(-\frac{1}{2}\right)$$

$$= \frac{29}{4}$$

$$b) i) x^2 - 4x = 4y$$

$$x^2 - 4x + 4 = 4y + 4$$

$$(x-2)^2 = 4(y+1)$$

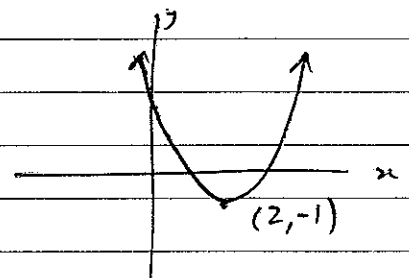
$$ii) \text{ In the form } (x-h)^2 = 4a(y-k)$$

$$4a = 4$$

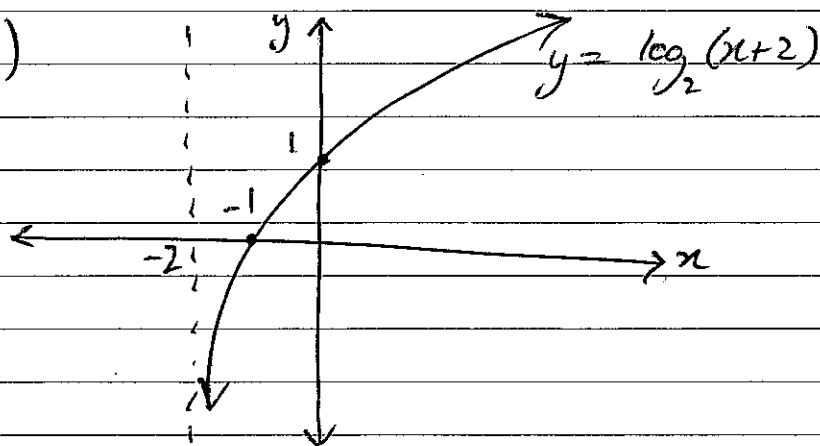
$$a = 1$$

Vertex $(2, -1)$
Focus $(2, 0)$

$$iii) y = -2$$



5) a) i)



ii) Domain: $x > -2$
Range: all real y .

b) $A(x-1)^2 + Bx + C \equiv x^2$
equate coefficients of x^2

$$A = 1$$

let $x = 0$

$$A(0-1)^2 + B(0) + C = (0)^2$$

$$A + C = 0$$

$$C = -A$$

$$C = -1$$

let $x = 1$

$$A(1-1)^2 + B(1) + C = (1)^2$$

$$B + C = 1$$

$$B + (-1) = 1$$

$$B = 2$$

c) $T_9 = 78$

$$T_{17} = 50$$

$$T_n = a + (n-1)d$$

$$a + 8d = 78 \quad \text{--- (1)}$$

$$a + 16d = 50 \quad \text{--- (2)}$$

$$\textcircled{2} - \textcircled{1}$$

$$8d = -28 \quad \text{--- (3)}$$

$$d = -3.5$$

sub ③ into ①

$$a + (-28) = 78$$

$$\underline{a = 106}$$

$$\text{ii) } S_n = \frac{n}{2} (2a + (n-1)d)$$

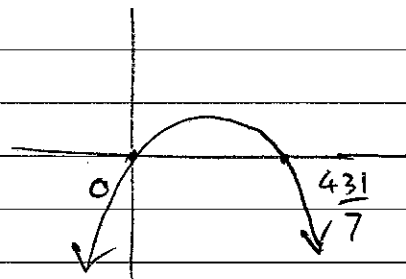
$$S_{40} = \frac{(40)}{2} (2(106) + ((40)-1)(-3.5))$$

$$= 1510$$

$$\text{iii) } S_n < 0$$

$$\frac{n}{2} (2(106) + (n-1)(-3.5)) < 0$$

$$\frac{n}{2} (215.5 - 3.5n) < 0$$



since n is a positive integer

$$n > \frac{431}{7}$$

$$n > 61.571428$$

$$\underline{n = 62}$$

$$e) a) f(x) = x^2 - x$$

$$f(x+h) = (x+h)^2 - (x+h) \\ = x^2 + 2xh + h^2 - x - h$$

OR

$$(x+h)(x+h-1)$$

$$b) 0.42 + 0.0042 + 0.000042 + \dots$$

$$S_{\infty} = \frac{a}{1-r}$$

$$S_{\infty} = \frac{\left(\frac{42}{100}\right)}{1 - \left(\frac{1}{100}\right)}$$

$$= \frac{14}{33}$$

OR

$$0.\dot{4}\dot{2} = 0.42 + 0.0042 + 0.000042 + \dots$$

$$\text{let } x = 0.4242\dots$$

$$100x = 42.4242\dots$$

$$99x = 42$$

$$x = \frac{42}{99}$$

$$x = \frac{14}{33}$$

$$c) i) f(-1) + f(2) = [-2] + [(2)^2 + 1] \\ = 3$$

$$ii) f(a^2+2) = (a^2+2)^2 + 1$$

Note: $a^2+2 > 2$

$$= a^4 + 2a^2 + 4 + 1$$

$$= a^4 + 2a^2 + 5$$

$$d) i) 4^x = 12(2)^x - 32$$

$$(2^2)^x - 12(2)^x + 32 = 0$$

$$(2^x)^2 - 12(2)^x + 32 = 0$$

$$\text{let } y = 2^x$$

$$y^2 - 12y + 32 = 0$$

$$\begin{array}{r|l} x & 32 \\ + & -12 \end{array}$$

$$(y-8)(y-4) = 0$$

$$y = 8 \quad \text{or} \quad y = 4$$

$$2^x = 8$$

$$2^x = 4$$

$$2^x = 2^3$$

$$2^x = 2^2$$

$$\therefore x = 3, 2$$

$$ii) 2 \log_5 3 = \log_5 x - \log_5 6$$

$$\log_5 3^2 = \log_5 \left(\frac{x}{6} \right)$$

$$\frac{x}{6} = 9$$

$$x = 54$$